

## Teacher Notes

This section provides teachers with a quick reference for each chapter: the intended learning outcomes, common misconceptions to watch for, suggested classroom activities, and remediation tips for learners who need extra support.

### Chapter 1: Large Numbers Around Us

**Learning Outcomes:** Students can read, write, compare and round large numbers in both the Indian and International numbering systems, and apply this to real data.

**Common Misconceptions:** Confusing crore or lakh with million or billion; errors in comma placement when switching between the two systems.

**Suggested Classroom Activities:** Use real population, budget or distance data from newspapers; build a place-value chart comparing both systems side by side.

**Remediation Tips:** Practise converting a few numbers between systems repeatedly using a labelled place-value chart before moving to word problems.

### Chapter 2: Arithmetic Expressions

**Learning Outcomes:** Students can apply BODMAS correctly, and form and evaluate real-life arithmetic expressions.

**Common Misconceptions:** Performing operations strictly left-to-right while ignoring precedence; forgetting that brackets change the order of evaluation.

**Suggested Classroom Activities:** Bill-making and shopping role play; expression-building games using digit and operation cards.

**Remediation Tips:** Practise one operation type at a time before mixing them; use colour-coding to mark each step of BODMAS as it is applied.

### Chapter 3: A Peek Beyond the Point

**Learning Outcomes:** Students understand decimal place value, convert between fractions and decimals, and perform decimal operations in real contexts.

**Common Misconceptions:** Misaligning decimal points during addition or subtraction; thinking more digits after the decimal point always means a bigger number.

**Suggested Classroom Activities:** Use decimal grids or base-ten blocks; calculate using real bills and price tags.

**Remediation Tips:** Align decimal points using grid paper, and compare numbers digit-by-digit starting from the left of the decimal point.

## Chapter 4: Expressions Using Letter-Numbers

**Learning Outcomes:** Students understand variables, and can form and evaluate algebraic expressions and identify terms, coefficients and constants.

**Common Misconceptions:** Treating letters as fixed labels rather than variable quantities; errors when combining like terms.

**Suggested Classroom Activities:** Real-life expression-building with changing quantities (ages, costs); substitution practice with number cards.

**Remediation Tips:** Use concrete substitution exercises before moving to symbolic simplification of expressions.

## Chapter 5: Parallel and Intersecting Lines

**Learning Outcomes:** Students can identify and calculate angles formed by intersecting lines and by a transversal cutting parallel lines.

**Common Misconceptions:** Confusing alternate and corresponding angles; assuming all angle pairs are equal regardless of whether the lines are actually parallel.

**Suggested Classroom Activities:** Use two rulers and a third ruler as a transversal to model angle pairs; measure real angles with a protractor.

**Remediation Tips:** Use colour-coded diagrams for each angle-pair type, with plenty of labelled-diagram practice before algebraic versions.

## Chapter 6: Number Play

**Learning Outcomes:** Students can apply divisibility rules, find prime factorisation, and use HCF and LCM to solve real grouping and scheduling problems.

**Common Misconceptions:** Confusing when to use HCF versus LCM; mistakenly thinking 1 is a prime number.

**Suggested Classroom Activities:** Factor-tree activities; real-life HCF and LCM word problems such as bell-ringing and packing.

**Remediation Tips:** Use factor trees and simple Venn diagrams; repeatedly practise distinguishing “share equally” (HCF) problems from “happen together” (LCM) problems.

## Chapter 7: A Tale of Three Intersecting Lines

**Learning Outcomes:** Students can classify triangles, and apply the angle sum property, the exterior angle theorem, and the triangle inequality.

**Common Misconceptions:** Believing any three lengths can form a triangle; forgetting the angle sum property when solving for an unknown angle.

**Suggested Classroom Activities:** Paper-triangle cutting and angle measuring; building real triangular structures with straws or sticks.

**Remediation Tips:** Use physical strips or straws to hands-on test the triangle inequality; repeated angle-sum practice with labelled diagrams.

## Chapter 8: Working with Fractions

**Learning Outcomes:** Students can perform all four operations on fractions and mixed numbers, and apply them to real-life measurement and rate problems.

**Common Misconceptions:** Adding or subtracting numerators and denominators directly without finding a common denominator; forgetting to flip the second fraction when dividing.

**Suggested Classroom Activities:** Recipe-scaling activities; fraction-strip manipulatives for visualising operations.

**Remediation Tips:** Practise finding common denominators systematically, and use visual fraction models for multiplication and division.

## Chapter 9: Operations on Integers

**Learning Outcomes:** Students can perform all four operations on integers and apply their properties in real-life contexts.

**Common Misconceptions:** Sign errors when multiplying or dividing two negative numbers; confusing subtraction of a negative with addition.

**Suggested Classroom Activities:** Number-line walking activities (physically stepping forward or backward); real temperature and elevation word problems.

**Remediation Tips:** Use number lines consistently; practise the “two negatives make a positive” rule with concrete examples like debts and temperature drops.

## Chapter 10: Fractions and Proportional Reasoning

**Learning Outcomes:** Students can simplify ratios, apply the unitary method, and calculate percentages and discounts in real contexts.

**Common Misconceptions:** Confusing the order of a ratio ( $a:b$  vs  $b:a$ ); errors converting between fractions, decimals and percentages.

**Suggested Classroom Activities:** Real shopping and discount calculations; recipe-scaling activities using ratios.

**Remediation Tips:** Practise the unitary method step-by-step (find the value of one unit first); use a fraction-decimal-percentage equivalence chart.

## Chapter 11: Finding the Unknown

**Learning Outcomes:** Students can solve linear equations in one variable using the balancing-and-transposition method, and translate word problems into equations.

**Common Misconceptions:** Forgetting to apply an operation to both sides of the equation; sign errors when transposing terms.

**Suggested Classroom Activities:** Balance-scale model demonstrations; word-problem to equation translation practice.

**Remediation Tips:** Use a physical or drawn balance-scale model; practise transposition rules and check solutions by substitution.

## Chapter 12: Congruent Figures

**Learning Outcomes:** Students understand congruence and can correctly apply the SSS, SAS, ASA and RHS criteria.

**Common Misconceptions:** Assuming equal angles alone (AAA) prove congruence; mismatching corresponding parts between the two triangles.

**Suggested Classroom Activities:** Paper-cutting and overlay activities to physically test congruence; discuss real-world examples like tiles and trusses.

**Remediation Tips:** Use tracing paper to overlay triangles, and carefully label corresponding vertices before applying any criterion.

## Chapter 13: Visualising Solid Shapes

**Learning Outcomes:** Students can differentiate 2D and 3D shapes, understand nets, and identify different views of solids.

**Common Misconceptions:** Confusing the number of faces with the number of surfaces visible at once; difficulty visualising a 3D solid from its 2D net.

**Suggested Classroom Activities:** Building solids from paper nets; view-drawing activities using cube arrangements.

**Remediation Tips:** Use physical 3D models and real packaging boxes, repeatedly unfolding and refolding them.

## Chapter 14: Comparing Quantities

**Learning Outcomes:** Students can calculate profit or loss percentage, discount, marked price and simple interest in real commercial situations.

**Common Misconceptions:** Calculating a percentage against the wrong base quantity (selling price instead of cost price, or vice versa); confusing discount percentage with profit percentage.

**Suggested Classroom Activities:** Shop or market role-play with marked prices and discounts; simple interest calculations using real bank rates.

**Remediation Tips:** Clearly identify the “base” (100

## Chapter 15: Data Handling

**Learning Outcomes:** Students can calculate mean, median, mode and range, and construct and interpret bar graphs from real data.

**Common Misconceptions:** Confusing mean with median; forgetting to arrange data in order before finding the median; miscounting the mode in tied situations.

**Suggested Classroom Activities:** Class survey and bar-graph construction; analysing real sports or exam-score data.

**Remediation Tips:** Practise systematically arranging data before finding the median; use simple repeated datasets to build comfort with each measure.

## Chapter 16: Symmetry

**Learning Outcomes:** Students can identify line and rotational symmetry, calculate the order and angle of rotation, and connect symmetry to natural and cultural patterns.

**Common Misconceptions:** Confusing the order of rotational symmetry with the number of lines of symmetry; assuming every symmetric figure has both types of symmetry.

**Suggested Classroom Activities:** Paper-folding and rotating cut-out activities; rangoli or mandala design activities.

**Remediation Tips:** Use tracing-paper overlays to physically test rotational symmetry, with clear examples distinguishing line versus rotational symmetry.

## Answer Key

Answers to all worksheets are provided below, in chapter order. For open-ended reflection and real-life application questions, and for any drawing/geometry activity, teachers should use their own judgement: these are marked as “Teacher Verification Required.”

### Worksheet 1: Large Numbers Around Us – Answer Key

1. Seven crore eighty-four lakh thirty-two thousand one hundred fifty-six
2. Ten million
3. 4,60,000
4. 32,40,006
5.  $12,34,567 < 12,43,567$
6. 8,45,000
7. 45,00,000 (45 lakh)
8. 3,80,000 km
9. 10,35,689
10.  $9,75,420 - 2,04,579 = 7,70,841$
11. Twelve crore forty-five lakh sixty-seven thousand eight hundred ninety; place value of 6 is 60,000
12. 2,50,00,000 (2 crore 50 lakh)
13. 34,65,789 is larger by 9,000
14. 78,00,000
15. 9,87,654; 9,86,754; 9,78,654; 9,76,845
16.  $60,000 \times 1,200 = 7,20,00,000$  (7 crore 20 lakh)
17. Rounded budget Rs. 10,00,00,000; each district  $\approx$  Rs. 40,00,000
18.  $1,00,00,000 = 1$  crore
19. Teacher Verification Required
20. Teacher Verification Required

*Reflection & Real-Life Questions: Teacher Verification Required.*

**Worksheet 2: Arithmetic Expressions – Answer Key**

1. 14
2. 20
3. 17
4. 5
5. The bracket  $(3 + 1)$
6. 40
7. 7
8.  $(2 + 3) \times 4 = 20$
9. 16
10. 90
11.  $50 + 20 \times 4 = \text{Rs. } 130$
12. 32
13. 10
14.  $2 \times (8 + 5) - 6 = 20$
15.  $3 + 4 \times 2 < (3 + 4) \times 2$  (11 vs 14)
16. e.g.  $(2 + 3) \times 4 = 20$
17.  $30 + 12 \times 7 + 15 = \text{Rs. } 129$
18. Correct value:  $15 - 3 \times 2 = 9$  (multiply before subtracting)
19.  $40 \times 3 + 20 \times 2 = \text{Rs. } 160$
20.  $100 \times 2 + 60 \times 3 + 50 = \text{Rs. } 430$

*Reflection & Real-Life Questions: Teacher Verification Required.*

**Worksheet 3: A Peek Beyond the Point – Answer Key**

1. 4 tenths and 5 hundredths
2. 0.7
3.  $\frac{1}{8}$
4. 5.68
5. 0.7

6. 6.23
7. 4.55
8. 0.45
9. 235 cm
10. 0.045, 0.405, 0.45, 0.5
11. 4.8
12. 8
13. Rs. 48.25
14. 1.625
15. 0.75 kg
16. 17 litres
17. Rs. 162.50
18.  $\frac{1}{3} = 0.333 \dots$  (repeats forever); approx 0.33
19. Teacher Verification Required
20. Teacher Verification Required

*Reflection & Real-Life Questions: Teacher Verification Required.*

### **Worksheet 4: Expressions Using Letter-Numbers – Answer Key**

1. 7
2.  $-9$
3.  $n + 5$
4. 19
5.  $3m - 6$
6.  $2l + 2b$
7.  $3p + 2q$
8. 24
9.  $4x, 3y, -7$
10.  $2x + 5$
11.  $4s$ ; 36 years



12.  $6x$
13.  $5a + 5b$
14.  $25n$ ; Rs. 375
15. 16
16.  $30 + 8d$ ; Rs. 126
17.  $2(x + 7) = 30 \Rightarrow x = 8$
18.  $2n + 5$
19.  $3a$ ; Rs. 360
20.  $3m$

*Reflection & Real-Life Questions: Teacher Verification Required.*

### **Worksheet 5: Parallel and Intersecting Lines – Answer Key**

1.  $180^\circ$
2.  $65^\circ$
3. They are equal
4. Angles on opposite sides of the transversal, between the two parallel lines
5.  $70^\circ$  (co-interior angles sum to  $180^\circ$ )
6.  $48^\circ$ ,  $132^\circ$ ,  $132^\circ$
7.  $75^\circ$
8. Angle BOD =  $130^\circ$  (vertically opposite); angle AOD =  $50^\circ$  (linear pair)
9.  $105^\circ$
10.  $90^\circ$  each
11. At each point:  $62^\circ$ ,  $118^\circ$ ,  $118^\circ$ ,  $62^\circ$  (all four angles repeat as  $62^\circ$  or  $118^\circ$ )
12. They are perpendicular to each other
13.  $55^\circ$  (corresponding angle)
14.  $3x + 2x = 180 \Rightarrow x = 36$ ; angles  $108^\circ$  and  $72^\circ$
15.  $60^\circ$ ,  $50^\circ$ ,  $70^\circ$ ,  $60^\circ$  (since the six angles must sum to  $360^\circ$  in vertically-opposite pairs)
16.  $65^\circ$  (corresponding angle)
17.  $2x + 10 = 3x - 15 \Rightarrow x = 25$ ; each angle =  $60^\circ$

18.  $71^\circ$  and  $109^\circ$
19. Teacher Verification Required
20. Teacher Verification Required

*Reflection & Real-Life Questions: Teacher Verification Required.*

## Worksheet 6: Number Play – Answer Key

1. Yes
2. 1, 2, 3, 4, 6, 9, 12, 18, 36
3. Yes (divisible by both 2 and 3)
4. 25
5.  $2^2 \times 3 \times 5$
6.  $2^5 \times 3$
7. 12
8. 36
9. Yes ( $144 = 8 \times 18 = 9 \times 16$ )
10. 53, 59, 61, 67
11. HCF = 6; LCM = 72
12. 36
13.  $2 + 3 + 5 + 7 + 11 = 28$
14. 5, 6, 9, 10, 15
15.  $2^3 \times 5^3$
16. LCM(12,18) = 36 minutes; at 9:36 am
17. HCF(48,60)=12 per row; 4 rows of roses, 5 rows of marigolds
18. LCM(3,4,5)=60; any multiple of 60 works, e.g. 1,020
19. Teacher Verification Required
20. Teacher Verification Required

*Reflection & Real-Life Questions: Teacher Verification Required.*

**Worksheet 7: A Tale of Three Intersecting Lines – Answer Key**

1.  $180^\circ$
2. Equilateral
3. Right-angled isosceles
4. No, since the two right angles alone would already total  $180^\circ$ , leaving no room for a third angle
5. Scalene
6.  $60^\circ$
7.  $60^\circ$
8. No, since  $3 + 4 = 7$ , which is less than 8
9.  $40^\circ$ ,  $60^\circ$ ,  $80^\circ$
10.  $55^\circ$
11.  $70^\circ$  each
12. Sum =  $125^\circ$ ; each =  $62.5^\circ$
13. No, since  $7 + 2 = 9$ , which is less than 10
14.  $37.5^\circ$ ,  $75^\circ$ ,  $67.5^\circ$
15. Equilateral; all three sides are equal
16.  $36^\circ$ ,  $60^\circ$ ,  $84^\circ$ ; all less than  $90^\circ$ , so it is acute
17. Between 7 cm and 23 cm
18. Interior at A =  $80^\circ$ ; at B =  $50^\circ$ ; at C =  $50^\circ$
19. Teacher Verification Required
20. Teacher Verification Required

*Reflection & Real-Life Questions: Teacher Verification Required.*

**Worksheet 8: Working with Fractions – Answer Key**

1.  $\frac{7}{10}$
2.  $\frac{5}{8}$
3.  $\frac{8}{15}$
4.  $\frac{9}{8} = 1\frac{1}{8}$

5.  $3\frac{2}{5}$
6.  $4\frac{1}{6}$
7.  $\frac{5}{8}$
8.  $\frac{1}{2}$
9.  $\frac{8}{5}$
10. 6
11. Used  $\frac{17}{3} = 5\frac{2}{3}$  m; remains  $2\frac{5}{6}$  m
12.  $1\frac{7}{8}$  cups
13.  $\frac{5}{3} = 1\frac{2}{3}$
14.  $\frac{3}{8}$
15. 9 pieces
16.  $2\frac{1}{2}$  days total;  $\frac{1}{5}$  of the job left after 2 days
17. 24 litres
18. 400 km
19. Teacher Verification Required
20.  $\frac{7}{20}$  of the day, which is 8.4 hours

*Reflection & Real-Life Questions: Teacher Verification Required.*

## Worksheet 9: Operations on Integers – Answer Key

1. -3
2. 9
3. -28
4. 6
5. 15
6. -160 m (160 m below sea level)
7.  $-7^{\circ}\text{C}$
8. 54
9. Rs. 300
10. 0

11.  $-120$
12. Floor 15
13. 24
14.  $15^{\circ}\text{C}$
15.  $-12$
16. 23 m below the surface
17. Both sides equal 0, so the property holds
18. Net profit of Rs. 6,000
19. Teacher Verification Required
20.  $-150 + 500 = +350$ , a net gain of Rs. 350

*Reflection & Real-Life Questions: Teacher Verification Required.*

## **Worksheet 10: Fractions and Proportional Reasoning – Answer Key**

1. 3:5
2. 30
3. 75%
4. Rs. 400
5. 2:3
6. 0.5 kg
7. 84
8. 300 km
9.  $\approx 28.6\%$
10. 20 boys, 25 girls
11. Discount Rs. 120; selling price Rs. 680
12. 16 days
13. 3 km
14. Milk 15 l, water 9 l
15. A: 4,800 votes; B: 3,200 votes

16. Scale factor 2.5; new shorter side 10 cm
17. Rs. 800, Rs. 1,200, Rs. 1,600
18. 30 litres needed; cost Rs. 3,060
19. Teacher Verification Required
20. Rs. 840

*Reflection & Real-Life Questions: Teacher Verification Required.*

### Worksheet 11: Finding the Unknown – Answer Key

1.  $x = 5$
2.  $x = 7$
3.  $x = 23$
4.  $x = 24$
5.  $x = 4$
6.  $x = 7$
7.  $x = 5$
8.  $x = 5$
9.  $x + 8 = 25 \Rightarrow x = 17$
10.  $x = 5$
11.  $x = 7$
12.  $x + 2x = 36 \Rightarrow x = 12$
13.  $x + 3x = 48 \Rightarrow x = 12$ ; son 12, father 36
14.  $x = 8$
15.  $x + (x + 1) + (x + 2) = 72 \Rightarrow x = 23$ ; numbers 23, 24, 25
16.  $2x - 5 = x + 3 \Rightarrow x = 8$
17.  $4x + 10 = 50 \Rightarrow x = 10$ ; width 10 cm, length 15 cm
18.  $x + (x + 12) = 48 \Rightarrow x = 18$ ; numbers 18 and 30
19.  $x + 5 = 20 \Rightarrow x = 15$
20.  $6x = 90 \Rightarrow x = 15$

*Reflection & Real-Life Questions: Teacher Verification Required.*

**Worksheet 12: Congruent Figures – Answer Key**

1. They are identical in shape and size
2. SSS, SAS, ASA, RHS
3. Yes
4.  $50^\circ$
5. Side-Angle-Side
6. SAS
7. RHS
8. Not necessarily congruent; equal angles alone only guarantee similarity, not congruence
9. ASA
10. 7 cm
11. Yes, by SAS
12. Yes, by ASA
13. Yes, by SSS
14. No, since two sides matching with an unequal included angle does not guarantee congruence
15. RHS
16. Yes; congruence does not depend on orientation, only on matching size and shape
17.  $AB=AC$ ,  $BD=DC$ , and  $AD$  is common to both, so by SSS they are congruent
18. Measure all three sides of the second section and check they match 4 m, 4 m, 5 m (SSS)
19. Teacher Verification Required
20. Teacher Verification Required

*Reflection & Real-Life Questions: Teacher Verification Required.*

**Worksheet 13: Visualising Solid Shapes – Answer Key**

1. Sphere
2. 5
3. A 2D flattened-out pattern that can be folded to form the solid
4. 5
5. Cylinder
6. 6
7. 1 flat face, 1 curved surface
8. Square
9. Rectangle
10. 6
11. Two faces of  $4 \times 3$  cm, two of  $4 \times 2$  cm, and two of  $3 \times 2$  cm
12. Square pyramid: 5 faces, 8 edges, 5 vertices
13. 125
14. Cylinder (open at both ends)
15. 5 faces, 9 edges, 6 vertices
16. 12 (the cubes along the edges, not corners)
17. Top view: circle; front and side views: rectangle
18. Teacher Verification Required
19. Teacher Verification Required
20. Teacher Verification Required

*Reflection & Real-Life Questions: Teacher Verification Required.*

**Worksheet 14: Comparing Quantities – Answer Key**

1. Rs. 100
2. Rs. 85
3. Rs. 200
4.  $SI = \frac{P \times R \times T}{100}$
5. Rs. 720



6. 20%
7. Rs. 405
8. Rs. 800
9. Rs. 1,320
10. Rs. 1,000
11. 20%
12. Discount Rs. 240; selling price Rs. 1,360
13. Interest Rs. 3,600; total amount Rs. 15,600
14. Rs. 1,000
15. CP of first = Rs. 400, CP of second = Rs. 625; total CP Rs. 1,025, total SP Rs. 1,000, overall loss of Rs. 25
16. 5 years
17. Profit of Rs. 80
18. Cost price  $\approx$  Rs. 1,500; discount = 15%
19. Rs. 1,050
20. Rs. 300

*Reflection & Real-Life Questions: Teacher Verification Required.*

## **Worksheet 15: Data Handling – Answer Key**

1. 8
2. 5
3. 5 (arranged: 2,4,5,7,9)
4. 18 ( $30 - 12$ )
5. Bars (rectangles) of varying heights
6. 56
7. 13.5 (arranged: 7,9,12,15,18,22; median of middle two 12 and 15)
8. Blue
9. 8 ( $33 - 25$ )
10. 35

11. Mean = 45; median = 52.5; mode = 60
12. 10.8
13. 34 (arranged: 12,19,23,34,45,50,67)
14. Mode = 78; it is the most frequently occurring score
15. Range = 30; mean = Rs. 56,667 (approx, i.e. 56.67 thousand)
16. 20
17. 240
18. Student A; A's range is 20, while B's range is 40, so A is more consistent
19. Teacher Verification Required
20. Teacher Verification Required

*Reflection & Real-Life Questions: Teacher Verification Required.*

## **Worksheet 16: Symmetry – Answer Key**

1. 6
2. No, but it has rotational symmetry
3. 3
4. The smallest angle through which the shape can be rotated to look identical to its original position
5. 1
6. 5
7.  $60^\circ$
8. No line symmetry; yes, rotational symmetry of order 2
9.  $90^\circ$
10. e.g. H, O, X, I
11. Order 8; angle  $45^\circ$
12.  $0^\circ$ ,  $72^\circ$ ,  $144^\circ$ ,  $216^\circ$ ,  $288^\circ$
13. 5 more (6 total)
14. Square: order 4; rectangle: order 2
15. 8 (a regular octagon)

16. Order 12; angle  $30^\circ$
17. Teacher Verification Required
18. Regular nonagon; each interior angle  $= 140^\circ$
19. Teacher Verification Required
20. Teacher Verification Required

*Reflection & Real-Life Questions: Teacher Verification Required.*